



# Computer Hardware & Maintenance

*Ground to Mastery — Complete Study Notes*

**CSC 121 | Year 1 Semester 2 | DCS | UICT Uganda**

Hardware · Software · CPU · Memory · Motherboard · Storage · Networks · Troubleshooting

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**Coursework 40% (Test 10% + Assignment 10% + Practical 20%) | Final Exam 60%**

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*Bravehart · TheRealistDevLab · 2025/2026*

## How to Use This Guide

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This guide takes you from absolute beginner to exam mastery for CSC 121 — Computer Hardware & Maintenance. Each unit is structured in three progressive levels:

### **BEGINNER**

BEGINNER — Understand what each component is and what it does. Perfect for when you first encounter a topic.

### **INTERMEDIATE**

INTERMEDIATE — Understand HOW things work together, key specifications, and how to identify components.

### **MASTERY**

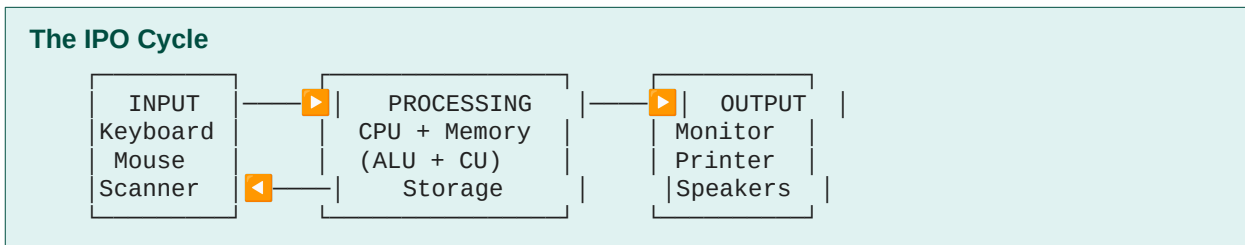
MASTERY — Troubleshoot problems, compare technologies, answer structured exam questions with confidence.

### **NOTE**

On Ubuntu/Linux you can verify hardware in real time using these terminal commands: `lscpu` → CPU info | `free -h` → RAM & swap | `lsblk` → storage | `lspci` → PCI devices | `lsusb` → USB devices | `sudo dmidecode` → full hardware info

# UNIT 1: Computer Hardware Fundamentals

A computer is an electronic device that accepts data (input), processes it according to stored instructions (programs), and produces output. It follows the IPO cycle.



## BEGINNER

A computer is like a factory: • INPUT = raw materials arriving (what you type or click) • PROCESSING = the factory floor (CPU doing the work) • OUTPUT = finished products (what appears on screen or prints) • STORAGE = the warehouse (hard drive saving everything)

## 1.1 — Key Definitions

| Term / Concept     | Definition / Explanation                                                                                      |
|--------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Hardware</b>    | Physical, tangible components you can touch — CPU, RAM, HDD, keyboard, monitor, cables                        |
| <b>Software</b>    | Instructions (programs) that tell hardware what to do — OS, applications, utilities                           |
| <b>Firmware</b>    | Software permanently stored in ROM — BIOS/UEFI — controls hardware at lowest level — cannot be easily changed |
| <b>Data</b>        | Raw facts and figures entered into the computer — unprocessed (e.g. "25", "Bravehart")                        |
| <b>Information</b> | Processed, meaningful data — the useful output (e.g. "Bravehart scored 25/30 = 83%")                          |

## 1.2 — Classification of Computers

| Term / Concept               | Definition / Explanation                                                                                                                              |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Supercomputer</b>         | Most powerful — billions of calculations per second — used for scientific research, weather forecasting, nuclear simulations. Example: Fugaku (Japan) |
| <b>Mainframe</b>             | Large, powerful — handles millions of transactions simultaneously — used by banks, airlines, governments. Example: IBM zSeries                        |
| <b>Minicomputer (Server)</b> | Medium-sized — serves multiple users at once — web servers, file servers, email servers. Example: Dell PowerEdge                                      |
| <b>Microcomputer (PC)</b>    | Personal computer — designed for one user — desktops, laptops, tablets. Most common type. Example: Dell, HP, MacBook                                  |

| Term / Concept           | Definition / Explanation                                                                                                                   |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Embedded Computer</b> | Built inside other devices — not visible as a computer — controls the device. Examples: microwave, car ECU, ATM, smart TV, washing machine |

### 1.3 — System Unit Components

| Term / Concept         | Definition / Explanation                                                                                      |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>CPU</b>             | Central Processing Unit — the brain — fetches, decodes, and executes instructions                             |
| <b>RAM</b>             | Random Access Memory — temporary working storage — data is lost when power is turned off                      |
| <b>ROM</b>             | Read-Only Memory — permanent — stores BIOS/firmware — data survives power-off                                 |
| <b>Motherboard</b>     | Main circuit board — connects all components — every other part plugs into it                                 |
| <b>PSU</b>             | Power Supply Unit — converts 220-240V AC mains power to DC voltages for components                            |
| <b>GPU</b>             | Graphics Processing Unit — renders images and video — can be integrated (on CPU) or dedicated (separate card) |
| <b>NIC</b>             | Network Interface Card — connects computer to a network — wired (Ethernet) or wireless (WiFi)                 |
| <b>Sound Card</b>      | Processes audio input and output — can be integrated on motherboard or separate card                          |
| <b>Expansion Slots</b> | PCIe and PCI slots on motherboard — used to add GPU, NIC, sound cards                                         |

### 1.4 — Input and Output Devices

#### Input Devices — bring data IN to the computer

- Keyboard — primary text and command input device
- Mouse / Touchpad — pointing, clicking, and selecting
- Scanner — converts physical documents and photos to digital format
- Microphone — audio input — voice recording, video calls
- Webcam — video input — video calls, security, streaming
- Barcode reader — reads barcodes at supermarkets and warehouses
- Biometric readers — fingerprint, retina scan — security and authentication
- Joystick / Gamepad — gaming input controllers

#### Output Devices — present results FROM the computer

- Monitor (VDU — Visual Display Unit) — visual output — LCD, LED, OLED technologies
- Printer — produces paper (hard copy) output — inkjet, laser, dot-matrix, thermal
- Speakers — audio output — plays sound, music, alerts

- Projector — large-screen visual output — classrooms, presentations
- Plotter — produces precise line drawings — engineering and architecture

### **MASTERY**

Exam question type: "State the difference between input and output devices and give two examples of each." Input devices bring data INTO the computer from the outside world. Output devices present processed information FROM the computer TO the user. Remember: Scanner = INPUT (it reads something IN). Printer = OUTPUT (it puts something OUT).

### **EXAM TIP**

Know the Von Neumann architecture: Input → Processing (CPU + Memory) → Output → Storage. The CPU fetches, decodes, and executes instructions in a cycle called the Fetch-Decode-Execute cycle. This is a very common exam question.

## UNIT 2: The Central Processing Unit (CPU)

The CPU is the brain of the computer. It is responsible for executing every instruction in every program. Without the CPU, the computer cannot function. All data must pass through the CPU to be processed.

### BEGINNER

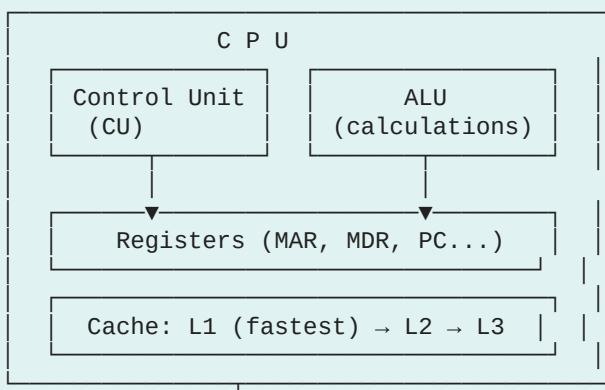
Think of the CPU like a chef in a kitchen:

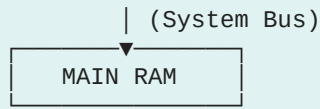
- The recipe (program) tells the chef what to do
- The chef reads one instruction at a time (fetch)
- Understands what it means (decode)
- Prepares/cooks it (execute)
- A faster chef (higher GHz) = faster cooking

### 2.1 — CPU Internal Components

| Term / Concept                       | Definition / Explanation                                                                                                                |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>ALU (Arithmetic Logic Unit)</b>   | Performs all arithmetic operations (add, subtract, multiply, divide) and logical comparisons (AND, OR, NOT, greater than, equal to)     |
| <b>CU (Control Unit)</b>             | Directs and coordinates all CPU activities — fetches instructions from RAM, decodes them, and tells the right component to execute them |
| <b>Registers</b>                     | Tiny, ultra-fast storage locations INSIDE the CPU — hold the data currently being processed — much faster than RAM                      |
| <b>L1 Cache</b>                      | Smallest and fastest cache — built directly into each CPU core — checked first                                                          |
| <b>L2 Cache</b>                      | Larger than L1 but slightly slower — still very fast — also per core or shared                                                          |
| <b>L3 Cache</b>                      | Largest cache level — shared between all cores — slower than L1/L2 but faster than RAM                                                  |
| <b>Program Counter (PC)</b>          | Register that holds the memory address of the NEXT instruction to be fetched                                                            |
| <b>Memory Address Register (MAR)</b> | Holds the address in memory that is currently being accessed                                                                            |
| <b>Memory Data Register (MDR)</b>    | Holds the data that has just been fetched from or is about to be written to memory                                                      |

#### CPU Internal Structure





## 2.2 — The Fetch-Decode-Execute Cycle

This cycle is how the CPU processes every single instruction. It runs billions of times per second.

1. **FETCH** — The Control Unit reads the address from the Program Counter and retrieves the next instruction from RAM into the MDR
2. **DECODE** — The Control Unit decodes (interprets) the instruction to understand what operation is needed
3. **EXECUTE** — The ALU or appropriate component carries out the operation — result stored in a register
4. **STORE** — Result written back to RAM if needed. Program Counter updated to point to next instruction

### ● INTERMEDIATE

The cycle repeats billions of times per second. A 3 GHz CPU performs 3 billion cycles per second. Modern CPUs can process multiple instructions simultaneously using pipelining — one instruction being fetched while another is decoded while another executes.

## 2.3 — CPU Performance Factors

| Term / Concept                    | Definition / Explanation                                                                                                                  |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Clock Speed</b>                | Number of cycles per second — measured in GHz (gigahertz). Higher GHz = faster processing. Example: 3.8 GHz = 3.8 billion cycles/second   |
| <b>Number of Cores</b>            | Each core is an independent processing unit. More cores = better multitasking. 2-core (dual), 4-core (quad), 8-core (octa) are common     |
| <b>Hyper-threading</b>            | Intel technology — each physical core acts as 2 logical cores — improves multitasking performance                                         |
| <b>Turbo Boost</b>                | Intel technology — CPU automatically increases clock speed above base speed when needed and when temperature allows                       |
| <b>Cache Size</b>                 | More cache = less time waiting for RAM. L1 cache miss → check L2 → check L3 → go to slower RAM                                            |
| <b>Socket Type</b>                | Physical connector between CPU and motherboard — MUST be compatible. Intel uses LGA (pins on motherboard), AMD uses AM4/AM5 (pins on CPU) |
| <b>TDP (Thermal Design Power)</b> | Maximum heat the CPU generates in watts — determines cooling solution needed                                                              |
| <b>Architecture (nm process)</b>  | Smaller manufacturing process (7nm, 5nm) = more transistors, less heat, lower power use                                                   |

## 2.4 — CPU Installation Steps

5. Ground yourself — attach ESD wrist strap before touching any components
6. Align the CPU — match the triangle/arrow marker on CPU corner to socket corner — NO FORCE needed
7. Use ZIF socket (Zero Insertion Force) — CPU drops in with zero force applied
8. Lower the lever or latch to lock CPU into the socket
9. Apply thermal paste — place a small rice-grain sized dot in centre of CPU top surface
10. Attach heatsink and fan — align mounting holes and press down until clips lock
11. Connect CPU fan power cable to CPU\_FAN header on motherboard

### **WARNING**

NEVER force a CPU into a socket. If it does not drop in freely, check alignment. Forcing it will bend or break pins permanently. Intel pins are on the motherboard (LGA) — AMD pins are on the CPU (AM4). One bent pin = motherboard or CPU is dead.

### **MASTERY**

CPU comparison question: "Compare Intel and AMD processors." Intel uses LGA sockets (pins on motherboard), AMD uses AM4/AM5 (pins on CPU). Both offer multi-core processors with cache hierarchy. Intel Turbo Boost ↔ AMD Precision Boost. AMD Ryzen generally offers more cores per price tier. Intel Core i9 ↔ AMD Ryzen 9 at the high end.

### **EXAM TIP**

The Fetch-Decode-Execute cycle is THE most important CPU concept. Draw it as a diagram if needed. Clock speed (GHz) and number of cores are the two main performance metrics. Know that cache hierarchy: L1 (fastest, smallest) → L2 → L3 (slowest, largest but still faster than RAM).



## UNIT 3: Memory Systems

Computer memory stores data and instructions. Different types of memory exist at different speeds, capacities, and costs. Understanding the memory hierarchy is fundamental to hardware maintenance.

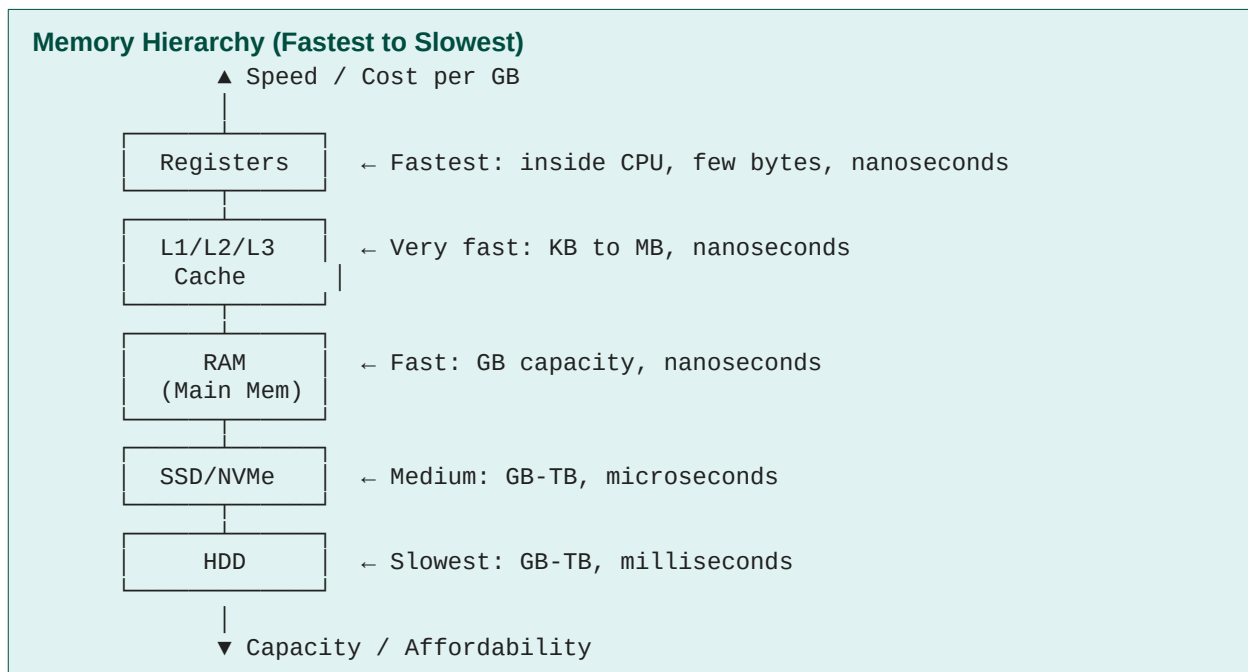
### BEGINNER

Memory is like a desk and filing cabinet:

- CPU Registers = the notes in your hand RIGHT NOW (fastest, tiny)
- Cache = the top of your desk within arm's reach (very fast, small)
- RAM = your full desk surface (fast, medium capacity)
- HDD/SSD = the filing cabinet (slower, huge capacity)

The closer to the CPU, the faster but the more expensive and smaller.

### 3.1 — Memory Hierarchy



### 3.2 — RAM (Random Access Memory)

| Term / Concept                  | Definition / Explanation                                                                          |
|---------------------------------|---------------------------------------------------------------------------------------------------|
| <b>SRAM (Static RAM)</b>        | Fastest RAM — used in CPU cache — holds data without refreshing — very expensive — small capacity |
| <b>DRAM (Dynamic RAM)</b>       | Main system memory — needs constant electrical refreshing to retain data — cheaper than SRAM      |
| <b>SDRAM (Synchronous DRAM)</b> | DRAM synchronized to CPU clock signal — older standard — largely obsolete                         |
| <b>DDR (Double Data Rate)</b>   | Transfers data on BOTH the rising AND falling edge of the clock — effectively doubles data rate   |
| <b>DDR4</b>                     | Current mainstream standard — 2133MHz to 3200MHz+ — 1.2V — most modern PCs use this               |
| <b>DDR5</b>                     | Newest generation — 4800MHz+ — lower voltage — higher                                             |

| Term / Concept          | Definition / Explanation                                                                                            |
|-------------------------|---------------------------------------------------------------------------------------------------------------------|
|                         | bandwidth — in latest PCs                                                                                           |
| <b>VRAM (Video RAM)</b> | Dedicated RAM on GPU — stores the frame buffer, textures, and rendering data                                        |
| <b>ECC RAM</b>          | Error-Correcting Code RAM — detects and corrects single-bit errors — used in servers — essential for data integrity |

#### **NOTE**

RAM must match your motherboard specifications exactly. DDR4 will not physically fit in a DDR5 slot. Check: type (DDR4/DDR5), speed (MHz), voltage (1.2V/1.1V), capacity per stick, number of sticks supported.

### 3.3 — ROM (Read-Only Memory)

| Term / Concept      | Definition / Explanation                                                                          |
|---------------------|---------------------------------------------------------------------------------------------------|
| <b>ROM</b>          | Read-Only Memory — factory-programmed — cannot be changed — stores fundamental boot code          |
| <b>PROM</b>         | Programmable ROM — blank from factory — user programs it ONCE — then permanent forever            |
| <b>EPROM</b>        | Erasable PROM — erased by exposure to UV light through a small window — then reprogrammable       |
| <b>EEPROM</b>       | Electrically Erasable PROM — erased using electrical signals — BIOS chips use this — no UV needed |
| <b>Flash Memory</b> | Modern EEPROM type — fast electrical erase and write — used in SSDs, USB drives, SD cards, phones |

### 3.4 — Virtual Memory

When RAM is full, the OS uses part of the hard disk as extra (virtual) memory. This area is called the swap file (Windows) or swap partition (Linux). Virtual memory is MUCH slower than real RAM but prevents crashes when RAM is exhausted.

```
# Check RAM and swap usage on Ubuntu
free -h

# Check swap partition details
swapon --show

# Check detailed RAM information
sudo dmidecode -t memory
```

#### **INTERMEDIATE**

Dual-channel memory: Installing two identical RAM sticks in the correct slots (usually same colour slots) allows the CPU to access both simultaneously — nearly doubling memory bandwidth. Always buy RAM in matched pairs for best performance.

#### **EXAM TIP**

KEY FACTS: RAM is volatile (data lost when power off). ROM is non-volatile (data survives power off). Cache is faster than RAM. HDD is non-volatile. Virtual memory uses HDD space — very slow compared to real RAM. Know the full hierarchy: Registers > L1 Cache > L2 Cache > L3 Cache > RAM > SSD > HDD.

## UNIT 4: Motherboard / System Board

The motherboard is the main printed circuit board (PCB) of the computer. Every other component either plugs directly into it or connects to it through cables. It determines what CPU, RAM, and expansion cards can be used.

### BEGINNER

Think of the motherboard as the city road network: • CPU = city centre (main hub) • RAM = nearby car parks (fast, close storage) • GPU = the airport (high-speed, specialised) • HDD/SSD = warehouses on the outskirts (slow, large storage) • Buses = the roads connecting everything Without roads (motherboard), nothing can communicate.

### 4.1 — Motherboard Form Factors

| Term / Concept       | Definition / Explanation                                                                             |
|----------------------|------------------------------------------------------------------------------------------------------|
| ATX                  | Standard: 305mm × 244mm — full-size desktop — most expansion slots — most features — standard choice |
| Micro-ATX (mATX)     | 244mm × 244mm — smaller — fewer PCIe slots — fits smaller cases — budget-friendly                    |
| Mini-ITX             | 170mm × 170mm — very compact — only 1 PCIe slot — for small form factor builds                       |
| E-ATX (Extended ATX) | 305mm × 330mm — larger than ATX — high-end workstations and servers — maximum expansion              |

### 4.2 — Key Motherboard Components

| Term / Concept        | Definition / Explanation                                                                                   |
|-----------------------|------------------------------------------------------------------------------------------------------------|
| Chipset               | Controls data flow between CPU, RAM, and peripherals. Determines what CPUs and features the board supports |
| CPU Socket            | Physical slot for the CPU. Must match CPU socket type exactly (LGA1700 for Intel, AM5 for AMD Ryzen)       |
| RAM Slots (DIMM)      | Slots for RAM modules — typically 2 or 4 slots — must use correct DDR generation                           |
| PCIe Slots            | High-speed expansion slots — x16 for GPU, x4/x1 for NVMe SSDs, NICs, sound cards                           |
| SATA Ports            | Connect HDDs, SSDs, and optical drives via SATA cable — typically 4-8 ports                                |
| M.2 Slot              | Connects NVMe SSDs directly to the board — very fast — no cables needed                                    |
| USB Headers           | Internal connectors for front-panel USB ports on the computer case                                         |
| 24-pin ATX Power      | Main power connector from PSU to motherboard                                                               |
| 4/8-pin EPS CPU Power | Additional CPU power connector — essential — computer won't boot without it                                |

| Term / Concept                | Definition / Explanation                                                                      |
|-------------------------------|-----------------------------------------------------------------------------------------------|
| <b>CMOS Battery</b>           | Small coin battery (CR2032) — powers CMOS memory to retain BIOS settings when computer is off |
| <b>Front Panel Headers</b>    | Connect power button, reset button, power LED, HDD LED from case to motherboard               |
| <b>POST Debug LEDs/Q-Code</b> | Shows diagnostic codes during startup — helps identify boot failures                          |

#### Motherboard Layout (Simplified)

|                        |                 |               |          |                |
|------------------------|-----------------|---------------|----------|----------------|
| [CPU Socket]           | [RAM Slot]      | [RAM Slot]    | [RAM]    | [RAM]          |
| [24-pin Power]         | [M.2 NVMe Slot] |               |          |                |
| [PCIe x16 – GPU]       |                 |               |          |                |
| [PCIe x1 – NIC/Sound]  |                 |               |          |                |
| [PCIe x1]              |                 |               |          |                |
| [SATA 1]               | [SATA 2]        | [SATA 3]      | [SATA 4] | [CMOS Battery] |
| [Front Panel Headers]  | [USB Headers]   | [Fan Headers] |          |                |
| I/O Panel (back of PC) |                 |               |          |                |
| [USB]                  | [USB]           | [HDMI]        | [DP]     | [Ethernet]     |
|                        |                 | [Audio]       | [PS/2]   |                |

## 4.3 — Computer Buses

| Term / Concept                    | Definition / Explanation                                                                                           |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Data Bus</b>                   | Carries actual data between CPU, RAM, and components — width is 32-bit or 64-bit in modern systems — bidirectional |
| <b>Address Bus</b>                | Carries memory addresses — tells system WHERE to read/write data — wider bus = more addressable RAM                |
| <b>Control Bus</b>                | Carries control signals — read/write commands, interrupt signals, clock signals — unidirectional                   |
| <b>PCIe (PCI Express)</b>         | Modern high-speed serial bus — replaces PCI — used for GPU (x16), NVMe SSD (x4), NICs                              |
| <b>USB (Universal Serial Bus)</b> | External device connection — USB 2.0 (480 Mbps), USB 3.0 (5 Gbps), USB 3.2 (20 Gbps)                               |
| <b>SATA</b>                       | Serial ATA — connects HDDs, SSDs, optical drives — SATA III = 6 Gbps                                               |

#### NOTE

The data bus is **BIDIRECTIONAL** (data flows both ways). The address bus is **UNIDIRECTIONAL** (CPU tells memory where to look — not the other way). This is a common MCQ trap.

## 4.4 — BIOS / UEFI and POST

| Term / Concept                   | Definition / Explanation                                                                                                                                 |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BIOS</b>                      | Basic Input/Output System — original firmware in ROM chip — first code CPU executes when powered on — limited to 16-bit mode                             |
| <b>UEFI</b>                      | Unified Extensible Firmware Interface — modern replacement for BIOS — 32/64-bit — faster boot — supports drives >2TB — graphical interface — Secure Boot |
| <b>CMOS</b>                      | Complementary Metal-Oxide-Semiconductor — small battery-backed memory — stores BIOS settings (date, time, boot order, CPU settings)                      |
| <b>POST (Power-On Self Test)</b> | Hardware diagnostic that runs every time computer starts — tests CPU, RAM, GPU, storage — beep codes indicate errors                                     |
| <b>Boot Order</b>                | BIOS/UEFI setting — determines which storage device to check first for OS — USB first for installations                                                  |
| <b>BIOS Entry Keys</b>           | Press DEL, F2, or F12 (depends on manufacturer) during the FIRST SECOND of startup to enter BIOS setup                                                   |

### Common POST Beep Codes

| Term / Concept                | Definition / Explanation                      |
|-------------------------------|-----------------------------------------------|
| <b>1 short beep</b>           | POST passed — normal startup — no errors      |
| <b>2 short beeps</b>          | POST error — check display/BIOS settings      |
| <b>1 long + 2 short beeps</b> | Display/video card error — reseal GPU         |
| <b>Continuous long beeps</b>  | RAM error — reseal or replace RAM modules     |
| <b>3 long beeps</b>           | Keyboard error — check keyboard connection    |
| <b>No beep at all</b>         | No power, no speaker, or dead motherboard/CPU |

#### MASTERY

If a computer won't boot, POST beep codes are your first diagnostic tool. Count the beeps carefully — the pattern identifies the problem. No speaker = install a PC speaker on the front panel headers to hear beeps.

#### EXAM TIP

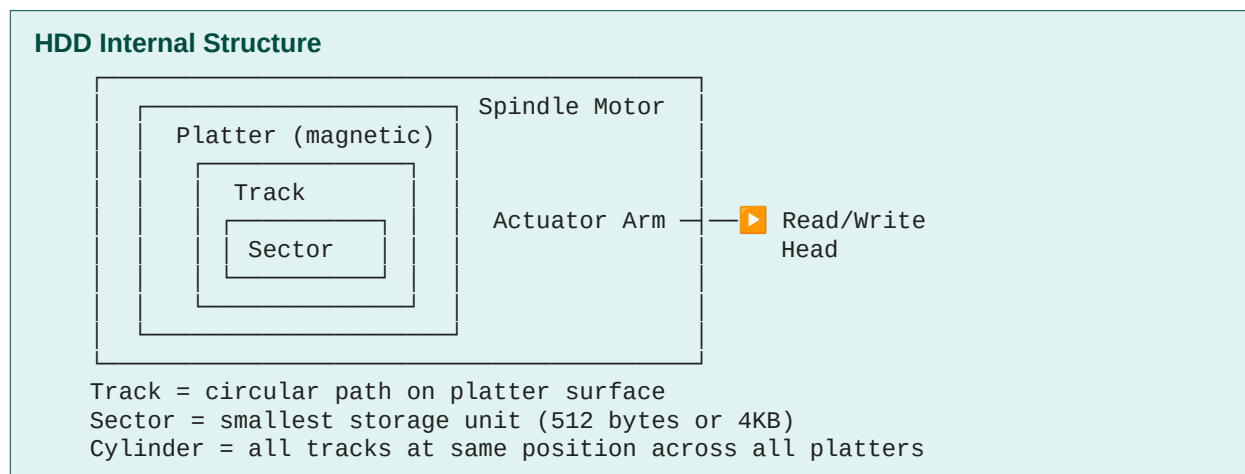
Common exam questions on motherboard: (1) What is POST and what does it do? (2) Why is the motherboard installed first and removed last? (3) State the difference between BIOS and UEFI. (4) What is the function of CMOS? Memorise these answers.

## UNIT 5: Storage Devices & Power Supply

Storage devices retain data permanently even when power is off. The Power Supply Unit converts mains electricity into safe DC voltages for all components.

### 5.1 — Storage Technologies Compared

| Term / Concept                 | Definition / Explanation                                                                                                |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>HDD (Hard Disk Drive)</b>   | Magnetic storage — spinning platters at 5400 or 7200 RPM — read/write heads — large capacity, low cost, fragile, slower |
| <b>SSD (Solid State Drive)</b> | Flash memory — no moving parts — fast, silent, shock-resistant, lighter — more expensive per GB than HDD                |
| <b>NVMe SSD</b>                | Next-gen SSD — uses PCIe bus — 5-10x faster than SATA SSD — plugs into M.2 slot — best performance available            |
| <b>Optical Drive</b>           | CD/DVD/Blu-ray — reads/writes optical discs using a laser — becoming obsolete — used for media, software, backups       |
| <b>USB Flash Drive</b>         | Portable flash storage — connects via USB — convenient — limited write cycles                                           |
| <b>SD Card</b>                 | Flash memory card — used in cameras, phones — sizes from microSD to full SD                                             |



### 5.2 — HDD Key Terms

| Term / Concept         | Definition / Explanation                                                                          |
|------------------------|---------------------------------------------------------------------------------------------------|
| <b>Platter</b>         | Magnetic disk inside HDD — one or more per drive — data stored magnetically on surface            |
| <b>Read/Write Head</b> | Reads and writes magnetic data — floats just above platter surface — one head per platter surface |
| <b>Spindle Motor</b>   | Motor that spins platters at constant speed — 5400 RPM (quiet/laptop) or 7200 RPM (performance)   |
| <b>Actuator Arm</b>    | Moves read/write heads to correct track position across the                                       |

| Term / Concept  | Definition / Explanation                                                                         |
|-----------------|--------------------------------------------------------------------------------------------------|
|                 | platter                                                                                          |
| <b>Track</b>    | Circular path on the platter surface — hundreds of concentric tracks per platter                 |
| <b>Sector</b>   | Smallest addressable unit on a platter — typically 512 bytes or 4096 bytes (4KB Advanced Format) |
| <b>Cluster</b>  | Group of consecutive sectors — minimum allocation unit used by file system — typically 4KB-64KB  |
| <b>Cylinder</b> | All tracks at the same position across ALL platters simultaneously                               |

### 5.3 — Drive Interfaces

| Term / Concept            | Definition / Explanation                                                                                                        |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>IDE/PATA</b>           | Parallel ATA — old interface — 40 or 80-pin ribbon cable — max 133 Mbps — supports 2 drives per cable (master/slave) — obsolete |
| <b>SATA</b>               | Serial ATA — current standard — thin 7-pin data cable — SATA III = 6 Gbps — one drive per cable — no jumpers needed             |
| <b>M.2 (SATA or NVMe)</b> | Form factor for SSDs — slots directly into motherboard — NVMe version uses PCIe for much faster speeds                          |
| <b>USB</b>                | For external drives — USB 3.0 = 5 Gbps, USB 3.2 = 20 Gbps — convenient for portable storage                                     |

### 5.4 — IDE vs SATA Comparison

| Feature                 | IDE/PATA vs SATA                                                     |
|-------------------------|----------------------------------------------------------------------|
| <b>Data cable width</b> | IDE: 40-50mm wide ribbon (80 wires)   SATA: 8mm thin 7-wire cable    |
| <b>Drives per cable</b> | IDE: 2 drives (master + slave)   SATA: 1 drive only                  |
| <b>Jumpers required</b> | IDE: YES — must set master/slave jumper   SATA: NO — not required    |
| <b>Speed</b>            | IDE: up to 133 Mbps   SATA III: up to 600 Mbps (6 Gbps)              |
| <b>Power connector</b>  | IDE: 4-pin Molex   SATA: 15-pin SATA power                           |
| <b>Modern use</b>       | IDE: Obsolete — not found in modern systems   SATA: Current standard |

### 5.5 — File Systems

| Term / Concept | Definition / Explanation                                                                                      |
|----------------|---------------------------------------------------------------------------------------------------------------|
| <b>FAT32</b>   | File Allocation Table 32 — Windows — max single file size 4GB — max partition 8TB — cross-platform compatible |



| Term / Concept | Definition / Explanation                                                                                               |
|----------------|------------------------------------------------------------------------------------------------------------------------|
| <b>NTFS</b>    | New Technology File System — Windows — supports files larger than 4GB — security permissions — encryption — journaling |
| <b>ext4</b>    | Fourth Extended Filesystem — Linux/Ubuntu default — journaling — efficient — supports large files and partitions       |
| <b>exFAT</b>   | Extended FAT — cross-platform (Windows/Mac/Linux) — used on USB drives and SD cards — no 4GB file limit                |
| <b>APFS</b>    | Apple File System — macOS and iOS — snapshots, encryption, space sharing                                               |

## 5.6 — Disk Management Operations

| Term / Concept         | Definition / Explanation                                                                                              |
|------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Formatting</b>      | Prepares a disk by creating a file system — sets up data structures — ERASES ALL DATA — cannot be undone easily       |
| <b>Partitioning</b>    | Divides one physical disk into multiple logical sections (C:, D:, etc.) — each partition can have its own file system |
| <b>Defragmentation</b> | HDD ONLY — rearranges fragmented files so they are stored contiguously — improves read speed — NEVER defrag an SSD    |
| <b>Disk Check</b>      | Scans for file system errors — CHKDSK (Windows) or fsck (Linux) — can mark and recover bad sectors                    |
| <b>Disk Cloning</b>    | Creates exact sector-by-sector copy of one drive to another — used for upgrades and backups                           |
| <b>RAID</b>            | Redundant Array of Independent Disks — multiple disks for speed (RAID 0), redundancy (RAID 1), or both (RAID 5)       |

## 5.7 — Power Supply Unit (PSU)

The PSU converts 220-240V AC from the mains socket into the stable DC voltages required by computer components. A failing PSU is a very common cause of system instability.

| Term / Concept             | Definition / Explanation                                                                      |
|----------------------------|-----------------------------------------------------------------------------------------------|
| <b>Function</b>            | Converts AC mains (220-240V Uganda) to DC voltages: +3.3V, +5V, +12V for different components |
| <b>+12V rail (yellow)</b>  | Powers CPU, GPU (graphics card), and motors in HDDs and cooling fans — highest demand         |
| <b>+5V rail (red)</b>      | Powers motherboard logic circuits, older drives, USB ports, optical drives                    |
| <b>+3.3V rail (orange)</b> | Powers RAM, modern motherboard components, PCI slots, M.2 devices                             |
| <b>Ground (black)</b>      | Reference voltage (0V) — all measurements are taken relative to ground                        |

| Term / Concept        | Definition / Explanation                                                                     |
|-----------------------|----------------------------------------------------------------------------------------------|
| <b>Wattage</b>        | Total power output — must exceed total system consumption with 20-30% headroom               |
| <b>80 PLUS Rating</b> | Efficiency standard — Bronze (82%), Silver (85%), Gold (87%), Platinum (90%), Titanium (94%) |

## PSU Connectors

| Term / Concept             | Definition / Explanation                                                      |
|----------------------------|-------------------------------------------------------------------------------|
| <b>24-pin ATX</b>          | Main power to motherboard — always required — large flat connector            |
| <b>4/8-pin EPS</b>         | Additional CPU power — connects near CPU socket — required for boot           |
| <b>6/8-pin PCIe</b>        | Power for discrete graphics cards (GPU) — needed for gaming/design GPUs       |
| <b>SATA Power (15-pin)</b> | Powers SATA HDDs, SSDs, and optical drives                                    |
| <b>Molex (4-pin)</b>       | Older connector — powers older drives, some fans, lighting — being phased out |



### PRACTICAL

PSU testing with a multimeter (power off, then use paperclip test):

1. Short the green wire (PS\_ON) to any black wire (ground) with a paperclip
2. Fan spins = PSU is alive and outputting power
3. Measure yellow wire to black = should read ~12V DC
4. Measure red wire to black = should read ~5V DC
5. Measure orange wire to black = should read ~3.3V DC

Values more than 5% off = PSU is degraded and should be replaced



### EXAM TIP

**Wattage:** Add up all component power requirements (CPU TDP + GPU TDP + RAM + drives + fans) then add 20-30% headroom. A system with 400W total draw needs a 500-600W PSU minimum. An undersized PSU causes random crashes, instability, and system damage.

## UNIT 6: Computer Software & Operating Systems

Software is the set of instructions that makes hardware perform useful tasks. Without software, hardware is just expensive metal and silicon.

### 6.1 — Classification of Software

| Term / Concept                          | Definition / Explanation                                                                                    |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>System Software</b>                  | Manages hardware resources and provides a platform for other software — OS, device drivers, BIOS, utilities |
| <b>Application Software</b>             | Performs specific tasks for end users — Word, Excel, Chrome, VLC, Photoshop, games                          |
| <b>Utility Software</b>                 | Performs maintenance and support tasks — antivirus, disk cleaner, file manager, backup tools, compression   |
| <b>Device Drivers</b>                   | Software that allows the OS to communicate with specific hardware — printer driver, GPU driver, NIC driver  |
| <b>Programming Language Translators</b> | Convert human-readable code to machine code — Assembler, Compiler, Interpreter                              |

### Language Translators

| Term / Concept     | Definition / Explanation                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------|
| <b>Assembler</b>   | Translates Assembly language (low-level, human-readable mnemonics) to machine code (binary)             |
| <b>Compiler</b>    | Translates entire high-level program (C, C++, Java) to machine code in one go — creates executable file |
| <b>Interpreter</b> | Translates and executes high-level code line by line — Python, JavaScript — slower but easier to debug  |

### 6.2 — Operating System Functions

12. Process Management — creates, schedules, and terminates programs — handles multitasking (running multiple programs simultaneously)
13. Memory Management — allocates RAM to running programs — manages virtual memory (swap) when RAM is full
14. File System Management — organises files into directories/folders — tracks file locations on disk
15. Device Management — communicates with hardware through drivers — manages I/O operations
16. User Interface — provides either CLI (Command Line Interface) or GUI (Graphical User Interface)
17. Security — user accounts and passwords, file permissions, encryption, firewall
18. Error Handling — detects, logs, and reports system errors — handles crashes gracefully

## 6.3 — Types of Operating Systems

| Term / Concept        | Definition / Explanation                                                                                     |
|-----------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Windows 11</b>     | Proprietary GUI OS by Microsoft — most common desktop OS — closed source — paid licence                      |
| <b>Linux / Ubuntu</b> | Open-source — free — many distributions (Ubuntu, Fedora, Debian) — highly customisable — what Bravehart uses |
| <b>macOS</b>          | Apple proprietary — based on Unix — only runs on Apple hardware — closed source                              |
| <b>DOS (Disk OS)</b>  | Command-line only — early Microsoft OS — precursor to Windows — concepts still used in CMD/PowerShell        |
| <b>UNIX</b>           | Multi-user, multi-tasking OS — basis for both Linux and macOS — used on servers                              |
| <b>Android</b>        | Mobile OS — Linux-based — Google — open source — most popular mobile OS globally                             |
| <b>iOS</b>            | Mobile OS — Apple — Unix-based — closed source — only on Apple devices                                       |

## 6.4 — Software Licensing

| Term / Concept              | Definition / Explanation                                                                                   |
|-----------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Open Source</b>          | Source code freely available — can view, modify, redistribute — Linux, LibreOffice, Firefox                |
| <b>Proprietary/Licensed</b> | Cannot distribute or modify source — must purchase licence — Windows, Adobe Photoshop, Microsoft Office    |
| <b>Freeware</b>             | Free to use but source code is closed — not open source — cannot be modified                               |
| <b>Shareware</b>            | Free trial period — must pay after trial expires — WinRAR is a classic example                             |
| <b>GNU/GPL</b>              | GNU General Public Licence — most common open source licence — requires derivatives to also be open source |

## 6.5 — OS Installation Steps

19. Set BIOS/UEFI boot order — set USB drive as first boot device
20. Boot from installation media — insert USB, restart, press boot menu key (F12, F8)
21. Select language, time zone, and keyboard layout
22. Choose installation type — Clean install (erases all) or Upgrade (preserves files)
23. Select and format the target partition — for clean install, delete all partitions and create new ones
24. Wait for file copying and installation to complete — may restart multiple times
25. Configure user account, username, password, region settings
26. Install drivers — GPU driver, NIC driver, audio driver — crucial for full functionality
27. Install Windows Updates or run: `sudo apt update && sudo apt upgrade` (Ubuntu)



**EXAM TIP**

Five OS functions must be memorised: Process management, Memory management, File system management, Device management, User interface. A sixth commonly asked: Security management.

## UNIT 7: Computer Troubleshooting & Maintenance

Troubleshooting is a systematic process for identifying, fixing, and preventing computer problems. A good technician always follows a logical process rather than guessing randomly.

### 7.1 — The 7-Step CompTIA Troubleshooting Process

28. Identify the problem — gather information from user, reproduce the issue, determine recent changes
29. Establish a theory — consider most likely causes based on symptoms (do not assume)
30. Test the theory — verify or eliminate the probable cause with a specific test
31. Establish an action plan — decide on the fix, obtain parts/tools needed, consider impact on user
32. Implement the solution — carry out the repair methodically
33. Verify full system functionality — test thoroughly — confirm with user that issue is resolved
34. Document findings — record the problem, root cause, and solution for future reference

#### BEGINNER

Simple troubleshooting rule: Always start with the SIMPLEST possible explanation. Computer won't turn on? → Check the power cable first. Screen is blank? → Check if monitor is on and cable is connected. Keyboard not working? → Try a different USB port. Do not replace expensive parts before checking the simple stuff.

### 7.2 — Common Hardware Problems & Solutions

| Term / Concept                       | Definition / Explanation                                                                                                                                     |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Computer won't power on</b>       | Check power cable and wall socket → check PSU rear switch is ON → test PSU with paperclip test → check power button cable on motherboard header              |
| <b>Screen blank / no display</b>     | Check monitor power and cable → try a different cable → reseal GPU → test with onboard video → listen for POST beeps                                         |
| <b>HDD not detected by BIOS</b>      | Check SATA data and power cables → try a different SATA port → check BIOS settings → drive may have failed (run diagnostics)                                 |
| <b>RAM failure (beeping)</b>         | Reseat RAM — remove, clean gold contacts with eraser, reinsert firmly → test one stick at a time → try different RAM slots                                   |
| <b>Overheating / random shutdown</b> | Check CPU temperature → clean dust from all fans and heatsinks with compressed air → check all fans are spinning → reapply thermal paste                     |
| <b>Computer very slow</b>            | Run Task Manager → check RAM usage (close unnecessary apps) → scan for malware → check HDD health with CrystalDiskInfo → consider defragmentation (HDD only) |
| <b>USB device not recognized</b>     | Try different USB port → reboot → update drivers via Device Manager → test device on another PC                                                              |
| <b>No sound</b>                      | Check speaker/headphone connection → check volume is not                                                                                                     |

| Term / Concept               | Definition / Explanation                                                                                  |
|------------------------------|-----------------------------------------------------------------------------------------------------------|
|                              | muted → check audio drivers in Device Manager → check correct output device is selected                   |
| <b>Printer not printing</b>  | Check power and cable/WiFi → check print queue for stuck jobs → reinstall driver → check ink/toner levels |
| <b>Computer freezes/BSOD</b> | Note the error code → check RAM (memtest86) → check HDD health → update drivers → check temperatures      |

## 7.3 — Types of Maintenance

| Term / Concept                | Definition / Explanation                                                                                                    |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Preventive Maintenance</b> | Regular tasks done BEFORE failures occur — cleaning, updates, backups, thermal paste replacement — cheapest in the long run |
| <b>Corrective Maintenance</b> | Fixing a problem AFTER it has occurred — replacing failed components, reinstalling OS — most expensive type                 |
| <b>Predictive Maintenance</b> | Monitoring warning signs to predict failure BEFORE it happens — SMART drive data, temperature logs, error logs              |
| <b>Adaptive Maintenance</b>   | Updating or modifying system to work with new hardware, software changes, or changed requirements                           |

## 7.4 — Preventive Maintenance Schedule

| Frequency               | Task                                                                                                                        |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Monthly</b>          | Check for OS and driver updates → run antivirus scan → backup important data → check disk usage                             |
| <b>Every 3-6 months</b> | Clean dust from all vents, fans, and heatsinks using compressed air → check all cable connections → review startup programs |
| <b>Every 1-2 years</b>  | Replace thermal paste on CPU → check HDD health (SMART data) → review and test UPS battery → update BIOS/UEFI if needed     |

## 7.5 — Troubleshooting Tools

| Term / Concept              | Definition / Explanation                                                                               |
|-----------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Multimeter</b>           | Measures voltage, current, resistance — tests PSU output voltages, cable continuity, component power   |
| <b>ESD Wrist Strap</b>      | Prevents static discharge that can permanently destroy components — always use when handling internals |
| <b>POST Diagnostic Card</b> | Plugs into PCIe/PCI slot — displays numeric POST codes to identify boot failures                       |
| <b>Cable Testers</b>        | Tests network cables for correct wiring — essential for network troubleshooting                        |
| <b>memtest86</b>            | Bootable RAM testing tool — runs extended tests for RAM                                                |

| Term / Concept         | Definition / Explanation                                                                       |
|------------------------|------------------------------------------------------------------------------------------------|
|                        | errors — leave running overnight for thorough test                                             |
| <b>CrystalDiskInfo</b> | Reads HDD/SSD SMART health data — shows reallocated sectors, temperature, power-on hours       |
| <b>CPU-Z / GPU-Z</b>   | Shows detailed information about CPU, RAM, and GPU — useful for verifying specifications       |
| <b>HWMonitor</b>       | Monitors temperatures, voltages, fan speeds in real time — essential for overheating diagnosis |



### PRACTICAL

Linux diagnostic commands — run these on your Ubuntu system: `lscpu` → CPU details (cores, speed, architecture) `free -h` → RAM and swap usage `lsblk` → all storage devices and partitions `lspci` → all PCI devices (GPU, NIC, sound) `lsusb` → all connected USB devices `df -h` → disk space per partition `sudo dmesg | tail -30` → recent kernel messages / hardware errors `sensors` → CPU and system temperatures `uptime` → system uptime and load averages



### MASTERY

When diagnosing any problem: 1) Talk to the user — when did it start? What changed? 2) Reproduce the problem. 3) Check simple things first (cables, power, connections). 4) Use diagnostic tools to confirm. 5) Fix one thing at a time. 6) Test after each change. 7) Document everything.



### EXAM TIP

Know all 7 troubleshooting steps in order. Know preventive vs corrective vs predictive maintenance. Know the ESD precautions — always in the exam. Know at least 5 common problems and their solutions.



## UNIT 8: DOS Commands & Networking Basics

MS-DOS (Microsoft Disk Operating System) is a command-line interface that predates Windows. DOS commands are still used in Windows Command Prompt and PowerShell. The concepts apply directly to Linux/Bash as well.

### 8.1 — Internal DOS Commands (built into COMMAND.COM)

| Term / Concept       | Definition / Explanation                                                          |
|----------------------|-----------------------------------------------------------------------------------|
| <b>DIR</b>           | List files and directories in current directory (equivalent to ls in Linux)       |
| <b>CD or CHDIR</b>   | Change directory: CD foldername (go in), CD.. (go up one level), CD\ (go to root) |
| <b>MD or MKDIR</b>   | Make directory: MD foldername — creates a new empty folder                        |
| <b>RD or RMDIR</b>   | Remove directory: must be empty first — cannot delete files inside                |
| <b>COPY</b>          | Copy files: COPY source.txt destination.txt — COPY *.txt D:\Backup\               |
| <b>DEL</b>           | Delete file: DEL filename.txt — irreversible from command prompt                  |
| <b>REN or RENAME</b> | Rename: REN oldname.txt newname.txt                                               |
| <b>TYPE</b>          | Display contents of text file: TYPE readme.txt (like cat in Linux)                |
| <b>CLS</b>           | Clear screen — like clear in Linux                                                |
| <b>DATE</b>          | Display or set the system date                                                    |
| <b>TIME</b>          | Display or set the system time                                                    |
| <b>VER</b>           | Display the current version of DOS/Windows                                        |
| <b>VOL</b>           | Display volume label and serial number of a disk                                  |
| <b>PATH</b>          | Display or set the executable search path                                         |

### 8.2 — External DOS Commands (separate .EXE or .COM files)

| Term / Concept | Definition / Explanation                                                        |
|----------------|---------------------------------------------------------------------------------|
| <b>FORMAT</b>  | Format a disk: FORMAT C: — WARNING: destroys ALL data on that drive permanently |
| <b>CHKDSK</b>  | Check disk for errors: CHKDSK C: /F (fix errors) /R (recover bad sectors)       |
| <b>FDISK</b>   | Partition disk — creates, deletes, and sets active partitions                   |
| <b>ATTRIB</b>  | Set file attributes: ATTRIB +R file.txt (read-only) +H (hidden) +S (system)     |
| <b>LABEL</b>   | Set disk volume label: LABEL C: SYSTEM                                          |

| Term / Concept | Definition / Explanation                                           |
|----------------|--------------------------------------------------------------------|
| <b>Sort</b>    | Sort output alphabetically: DIR   SORT — pipe output through SORT  |
| <b>Move</b>    | Move a file to different directory: MOVE file.txt D:\Documents\    |
| <b>XCOPY</b>   | Advanced copy with subdirectories: XCOPY C:\folder D:\backup /S /E |
| <b>Edit</b>    | Text editor for DOS — basic text editing from command line         |
| <b>SYS</b>     | Transfer OS system files to make a disk bootable                   |
| <b>HELP</b>    | Show help for any command: HELP CHKDSK                             |

#### **NOTE**

**KEY DIFFERENCE:** Internal commands are always available (part of COMMAND.COM). External commands require the .EXE file to be present on disk. If you get "Bad command or file name" for an external command, the .EXE file cannot be found in the PATH.

## 8.3 — Networking Basics

| Term / Concept                      | Definition / Explanation                                                                                        |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>LAN (Local Area Network)</b>     | Connects devices within a small area — home, office, school — typically 10Mbps to 10Gbps speeds                 |
| <b>WAN (Wide Area Network)</b>      | Connects devices over large geographic areas — the Internet is the world's largest WAN                          |
| <b>Router</b>                       | Directs data packets between networks — connects a LAN to the Internet — assigns IP addresses via DHCP          |
| <b>Switch</b>                       | Connects multiple devices within a LAN — smarter than a hub — directs data only to intended recipient           |
| <b>NIC (Network Interface Card)</b> | Hardware in each computer that enables network connection — has a unique MAC address                            |
| <b>IP Address</b>                   | Unique numerical address for each device on a network — IPv4: 192.168.1.5 — IPv6: longer hexadecimal            |
| <b>MAC Address</b>                  | Unique hardware identifier burned into every NIC at manufacture — format: XX:XX:XX:XX:XX:XX — cannot be changed |
| <b>DHCP</b>                         | Dynamic Host Configuration Protocol — server that automatically assigns IP addresses to devices on a network    |
| <b>DNS</b>                          | Domain Name System — translates domain names (google.com) to IP addresses (142.250.80.46)                       |

### Network Cable Categories

| Term / Concept | Definition / Explanation                                                 |
|----------------|--------------------------------------------------------------------------|
| <b>Cat5e</b>   | Supports up to 1 Gbps over 100 metres — most common in homes and offices |

| Term / Concept     | Definition / Explanation                                                                                    |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Cat6</b>        | Supports up to 10 Gbps over shorter distances — better shielding than Cat5e — common in modern offices      |
| <b>Cat6a</b>       | 10 Gbps over full 100 metres — used in professional/enterprise installations                                |
| <b>Fibre Optic</b> | Uses light pulses instead of electrical signals — very high speed — immune to interference — long distances |

## WiFi Standards

| Term / Concept           | Definition / Explanation                                                |
|--------------------------|-------------------------------------------------------------------------|
| <b>802.11n (WiFi 4)</b>  | Up to 600 Mbps — 2.4 GHz and 5 GHz — common in older devices            |
| <b>802.11ac (WiFi 5)</b> | Up to 3.5 Gbps — primarily 5 GHz — current mainstream standard          |
| <b>802.11ax (WiFi 6)</b> | Up to 9.6 Gbps — 2.4 GHz, 5 GHz, 6 GHz — better in crowded environments |

### EXAM TIP

Know internal vs external DOS commands. Know networking basics: LAN vs WAN, router vs switch, IP vs MAC address. WiFi standard numbers: 802.11n = WiFi 4, 802.11ac = WiFi 5, 802.11ax = WiFi 6. Higher number = faster.

## SECTION A: Multiple Choice Questions — Practice & Answers

---

40 Marks | 2 marks per question | Choose the best answer

**Q1.** Which of the following is NOT an input device?

- A. Keyboard
- B. Scanner
- C. Monitor**
- D. Barcode reader

**Answer:** C. Monitor

**Explanation:** Monitor is an *OUTPUT* device — it displays (outputs) information to the user. Keyboard, scanner, and barcode reader all bring data *IN* to the computer.

**Q2.** A light-sensitive device that converts printed text and images into digital form is called a:

- A. Plotter
- B. Printer
- C. Scanner**
- D. OMR

**Answer:** C. Scanner

**Explanation:** A scanner digitises physical documents by sensing light reflected from the document surface and converting it to digital data.

**Q3.** Information travels between components on the motherboard through:

- A. CMOS
- B. Buses**
- C. Bays
- D. Flash memory

**Answer:** B. Buses

**Explanation:** Buses are electronic pathways (circuits) on the motherboard that carry data, addresses, and control signals between components.

**Q4.** The data bus is \_\_\_\_\_ directional.

- A. Single
- B. Multi
- C. Bi**
- D. None of the above

**Answer:** C. Bi

**Explanation:** The data bus is *BIDIRECTIONAL* — data can travel both *TO* and *FROM* the CPU. The address bus is *unidirectional* (CPU to memory only).

**Q5.** Which is known as the translator for low-level Assembly language?

- A. Compiler
- B. Linker
- C. Loader
- D. Assembler**

**Answer:** D. Assembler

**Explanation:** An Assembler translates Assembly language (human-readable mnemonics like *MOV*, *ADD*) into machine code (binary 0s and 1s).

**Q6.** Which command divides the surface of a blank disk into sectors and assigns unique addresses?

- A. VER

**B. FORMAT**

C. FAT

D. CHKDSK

**Answer: B. FORMAT**

**Explanation:** *FORMAT creates the file system structure on a disk — sets up tracks, sectors, and the file allocation table. WARNING: it erases all data.*

**Q7.** The MOST common reason for a computer to overheat is:

A. Insufficient RAM

B. Outdated software

**C. Blocked ventilation or dust buildup**

D. Weak internet connection

**Answer: C. Blocked ventilation or dust buildup**

**Explanation:** *Dust accumulates on heatsink fins and fan blades over time, blocking airflow and reducing heat dissipation. Clean with compressed air every 3-6 months.*

**Q8.** When troubleshooting a computer that won't boot, the FIRST step should be:

A. Reinstall the operating system

B. Upgrade hardware

**C. Check for loose cables or faulty connections**

D. Run a virus scan

**Answer: C. Check for loose cables or faulty connections**

**Explanation:** *Always check the simplest possible cause first. A loose power cable is free to fix. OS reinstallation destroys data and should be a last resort.*

**Q9.** To safely work inside a computer, you should:

A. Work while it's powered on

B. Use any available screwdriver

**C. Ground yourself to prevent ESD**

D. Leave old components inside

**Answer: C. Ground yourself to prevent ESD**

**Explanation:** *ESD (Electrostatic Discharge) from your body can permanently destroy components. Use an anti-static wrist strap or touch unpainted metal chassis before touching components.*

**Q10.** Which tool is NOT typically used for computer hardware maintenance?

A. Multimeter

B. Anti-static wrist strap

**C. Power drill**

D. Thermal paste

**Answer: C. Power drill**

**Explanation:** *A power drill has no place in computer maintenance. The other three are all legitimate hardware maintenance tools.*

**Q11.** RAM is volatile because:

A. It runs too fast to save data

**B. It loses all data when power is removed**

C. It only stores a small amount

D. It is located inside the CPU

**Answer: B. It loses all data when power is removed**

**Explanation:** *Volatile memory requires constant electrical power to maintain its stored data. The instant power is removed, all data in RAM disappears completely.*

**Q12.** Which PSU connector provides the MAIN power to the motherboard?

A. 4-pin Molex

B. SATA 15-pin

**C. 24-pin ATX**

D. 6-pin PCIe

**Answer: C. 24-pin ATX**

**Explanation:** The 24-pin ATX connector is the main motherboard power connector — the large flat connector on the right side of the motherboard.

**Q13.** UEFI is an improvement over BIOS because:

A. It costs more

**B. It supports drives larger than 2TB and has faster boot times**

C. It only works on AMD processors

D. It replaces the operating system

**Answer: B. It supports drives larger than 2TB and has faster boot times**

**Explanation:** UEFI supports GPT partition tables (drives over 2TB), 32/64-bit mode (BIOS was 16-bit), graphical interface, network capabilities, and faster boot with Secure Boot.

**Q14.** The Fetch-Decode-Execute cycle is performed by:

A. The hard disk

B. The RAM

**C. The CPU**

D. The GPU

**Answer: C. The CPU**

**Explanation:** The CPU's Control Unit fetches instructions from RAM, decodes them, and the ALU executes them. This cycle is the fundamental operation of all CPUs.

**Q15.** Which file system does Ubuntu Linux use by default?

A. NTFS

B. FAT32

**C. ext4**

D. exFAT

**Answer: C. ext4**

**Explanation:** ext4 (Fourth Extended Filesystem) is the default filesystem for Ubuntu and most Linux distributions. Windows uses NTFS, USB drives typically use exFAT or FAT32.

**Q16.** A symptom of a failing hard drive might include:

A. Blurry screen resolution

B. Inability to connect to WiFi

**C. Frequent freezing or unexpected shutdowns**

D. Difficulty printing documents

**Answer: C. Frequent freezing or unexpected shutdowns**

**Explanation:** HDD failure causes file system errors which result in freezing, crashes, Blue Screen of Death (BSOD), and eventually complete failure to boot.

**Q17.** Which DOS command lists files in the current directory?

A. CD

**B. DIR**

C. MD

D. TYPE

**Answer: B. DIR**

**Explanation:** DIR (Directory) lists all files and subdirectories in the current location — equivalent to ls in Linux.

**Q18.** What is the purpose of thermal paste?

A. To attach the CPU to the socket

**B. To conduct heat from CPU to heatsink efficiently**

- C. To protect pins from bending
- D. To prevent CPU from overclocking

**Answer:** B. To conduct heat from CPU to heatsink efficiently

**Explanation:** Thermal paste fills microscopic air gaps between the CPU surface and heatsink base, dramatically improving thermal conductivity and heat transfer.

**Q19.** Which type of maintenance is done REGULARLY to PREVENT failures?

- A. Corrective maintenance
- B. Predictive maintenance
- C. Preventive maintenance**
- D. Adaptive maintenance

**Answer:** C. Preventive maintenance

**Explanation:** Preventive maintenance is done on a schedule BEFORE problems occur — cleaning, updates, backups. Corrective is done AFTER failure. Predictive monitors for warning signs.

**Q20.** When cleaning inside a computer, the correct tool to use is:

- A. A vacuum cleaner with high suction
- B. Water directly on components
- C. Compressed air to blow out dust**
- D. A damp cloth on circuit boards

**Answer:** C. Compressed air to blow out dust

**Explanation:** Compressed air is safe for electronics — it removes dust without touching components, without moisture, and without generating static electricity.

## SECTION B: Structured Questions — Answer Any THREE

Answer ANY THREE questions. Each question carries 20 marks. Show all reasoning clearly.

### Question 1

20

Answer all parts of this question.

(a) (i) **4 marks** Explain the purpose of an input device.

**Model Answer:** An input device is any hardware component that allows a user to enter data and instructions into a computer system. Input devices convert actions (keystrokes, mouse movements) or physical media (documents, audio) into digital signals the CPU can process. Examples include keyboard, mouse, scanner, microphone, and barcode reader.

(a) (ii) **4 marks** Explain the purpose of an output device.

**Model Answer:** An output device is any hardware component that presents processed information from the computer to the user. Output devices convert digital signals from the computer into a human-perceivable form. Examples include monitor (visual), printer (paper copy), speakers (audio), and projector (large-screen visual).

(a) (iii) **4 marks** Explain the purpose of main memory.

**Model Answer:** Main memory (RAM) is the computer's temporary working space where the CPU stores data and program instructions that are currently being processed. It is volatile (data lost when power off), fast access (nanoseconds), and directly affects how many programs can run simultaneously. More RAM = better multitasking.

(b) **4 marks** Explain the difference between application software and system software.

**Model Answer:** System software manages and controls computer hardware — it provides a platform for running other software. Examples: Windows, Ubuntu Linux, device drivers. Application software performs specific tasks for end users and runs ON TOP of the system software. Examples: Microsoft Word, Chrome, VLC media player, Photoshop.

(c) **4 marks** Give two examples of application software and two examples of system software.

**Model Answer:** Application software: Microsoft Word (word processing), Google Chrome (web browser). System software: Ubuntu Linux / Windows 11 (operating systems), GPU device driver (hardware driver).

### Question 2

20

Answer all parts of this question.

(a) **4 marks** Explain the meaning of an operating system.

**Model Answer:** An operating system (OS) is the most fundamental system software that manages all computer hardware resources and provides services to application programs. It acts as an interface between the user and the hardware, coordinates all activities of the computer, and ensures different programs can run without conflicting with each other.

(b) **6 marks** Explain three phases taken during OS installation.

**Model Answer:** 1. Preparation Phase: Set BIOS/UEFI boot order to boot from installation media (USB). Partition and format the target hard drive. Back up any existing data. 2. Installation/Copying Phase: OS installation files are copied from the installation media to the hard drive. The system may restart several times during this process. 3. Configuration Phase: User account creation, region and language settings, network configuration, and initial driver installation. Windows Update or apt update is run to apply latest patches.

(c) **10 marks** Discuss any five functions performed by an operating system.

**Model Answer:** 1. Process Management: The OS creates, schedules, and terminates processes. It manages multitasking by allocating CPU time to each running program using scheduling algorithms. 2. Memory Management: Allocates RAM to running programs, tracks which memory is in use, and manages virtual memory (swap) when physical RAM is insufficient. 3. File System Management: Organises files into directories and folders, tracks file locations on storage devices, manages file permissions, and handles read/write operations. 4. Device Management: Communicates with hardware devices through device drivers,



handles input/output operations, and manages conflicts between devices. 5. User Interface: Provides either a Command Line Interface (CLI) for text-based interaction or a Graphical User Interface (GUI) for mouse-based interaction.

### Question 3

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Answer all parts of this question.

**(a) 10 marks** State five factors to consider before purchasing application software.

**Model Answer:** 1. *Functionality:* Does the software perform the specific tasks you need? Check features against requirements. 2. *Compatibility:* Will it run on your operating system (Windows/Linux/macOS) and meet minimum hardware requirements (CPU, RAM, storage)? 3. *Cost and Licensing:* Is it within budget? Consider purchase price, annual subscription, or per-user licensing. Also consider ongoing maintenance costs. 4. *Ease of Use:* Is the interface user-friendly? Is there good documentation, tutorials, and technical support available? 5. *Security:* Does the software come from a reputable vendor? Does it have a history of vulnerabilities or data breaches? Are regular security updates provided?

**(b) 10 marks** Explain why the motherboard is the first element installed during assembly but the last removed during disassembly.

**Model Answer:** The motherboard is installed **FIRST** during assembly because: (1) It is the backbone/foundation of the entire system — all other components connect to it directly or through cables. (2) It determines the physical layout and positioning of all other components in the case. (3) Installing it first provides a stable platform and reference point for mounting the CPU, RAM, GPU, and storage devices. (4) Screw mounting holes and standoffs must be aligned with the motherboard before anything else can be placed. The motherboard is removed **LAST** during disassembly because: (1) All other components (CPU, RAM, GPU, expansion cards) must be removed before the motherboard can be accessed. (2) Power connectors (24-pin, CPU 8-pin), data cables (SATA), and front panel headers all connect to the motherboard from outside components. (3) Removing it last ensures systematic, organised disassembly that prevents damage. (4) The GPU occupies the PCIe slot, RAM occupies DIMM slots, and drives connect via SATA/M.2 — all must come off first.

### Question 4

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You are a computer technician at a repair shop on Kampala Road. A customer brings in their desktop PC complaining it is making a loud whirring noise.

**(a) 3 marks** State three potential causes of the loud whirring noise.

**Model Answer:** 1. A failing cooling fan (CPU fan, case fan, or PSU fan) with worn bearings — produces a grinding or rattling whir. 2. A failing hard disk drive with mechanical problems — spindle motor or read/write head issues produce distinctive clicking/whirring. 3. A loose cable or debris that has fallen into a spinning fan blade — produces intermittent or continuous rattling noise.

**(b) 7 marks** Explain how you would diagnose the problem step by step.

**Model Answer:** 1. Listen carefully with case closed — try to determine if noise is coming from front, back, side, or top. 2. Power off, unplug, ground yourself with ESD strap, open the case. 3. Visual inspection — check for dust buildup, loose cables near fan blades, foreign objects. 4. Power on with case open — observe fans — identify which is making noise. 5. Gently stop each fan temporarily with a non-conductive tool to isolate the noisy one. 6. For HDD — run CrystalDiskInfo or a SMART test to check drive health. 7. Document which component is the source and assess whether repair or replacement is needed.

**(c) 10 marks** Describe five possible solutions depending on your findings.

**Model Answer:** 1. Clean all dust: Use compressed air to thoroughly blow out dust from CPU heatsink, case fans, GPU heatsink, and PSU fan. Often resolves the problem completely. 2. Replace the failing fan: If a specific fan has failed or has worn bearings, remove and replace it with a compatible replacement. Match voltage, connector type, and size. 3. Remove debris: If a loose cable or foreign object is contacting a fan blade, carefully reposition cables using cable ties and remove any debris safely. 4. Replace failing HDD: If diagnostics show the HDD has mechanical issues (bad sectors, reallocated sectors, clicking sounds), immediately back up all data and replace the drive. 5. Reseat and secure components: Tighten any loose screws, reseat the GPU and RAM, and use cable management to prevent cables from vibrating or contacting fans. After each fix, test by powering on to verify the noise is resolved.

## Question 5

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A student brings in her laptop complaining it is shutting down unexpectedly and the fan is running very loudly all the time.

**(a) 8 marks** Describe four initial troubleshooting steps to diagnose the problem.

**Model Answer:** 1. Check temperatures using monitoring software: Install and run HWMonitor or use sensors in Linux to check CPU and GPU temperatures. Normal CPU temperature should be below 90°C under load. If temperatures are 95-100°C+, thermal throttling and shutdown protection is triggering. 2. Observe fan behaviour: Determine if the fan runs at maximum speed continuously (indicates the system is always too hot) or only during specific tasks (indicates software issue). 3. Check ventilation physically: Inspect the laptop vents on the bottom and sides. If covered with dust or debris, airflow is blocked. Also check if the laptop is being used on soft surfaces (bed, cushion) that block vents. 4. Check Windows/Linux power settings: Verify power management is not set to maximum performance mode which keeps CPU at 100% — this generates more heat. Also check if any processes are using high CPU (Task Manager / htop).

**(b) 8 marks** Based on findings, explain how you would describe the issue to the student and recommend a solution.

**Model Answer:** Explanation to student: "Your laptop is shutting down to protect itself from heat damage. Inside your laptop, the CPU generates a lot of heat when working. A fan and heatsink normally carry this heat away through vents. However, over time, dust builds up inside and blocks the airflow — like a blocked air vent. Your fan is working as hard as it can (that's the loud noise) but cannot remove heat fast enough. When the temperature gets dangerously high, the laptop automatically shuts down to prevent permanent damage." Recommendations: (1) We will open the laptop and use compressed air to clean all dust from the heatsink fins, fan blades, and all vents. (2) We will check if the thermal paste on the CPU has dried out and replace it if needed — this is common on laptops over 2 years old. (3) Always use the laptop on hard, flat surfaces — never on beds or cushions that block vents. (4) Consider a laptop cooling pad for heavy use. (5) Bring it for cleaning every 12 months.

**(c) 4 marks** After cleaning, temperatures are normal. What do you do next?

**Model Answer:** 1. Run a stress test (Prime95 or Cinebench) to verify temperatures remain stable under heavy load for 15-30 minutes. 2. Return the laptop to the student with written documentation of what was done, parts replaced (if any), and preventive care advice. 3. Advise on prevention: use on hard surfaces, annual cleaning, avoid dusty environments. 4. Schedule a follow-up in 6-12 months or advise student to return if problem recurs.

## Quick Reference — Key Facts for the Exam

### Assessment Breakdown

| Component            | Weight                           |
|----------------------|----------------------------------|
| Test                 | 10%                              |
| Assignment           | 10%                              |
| Practical (Lab work) | 20%                              |
| Final Examination    | 60%                              |
| TOTAL                | 100% — Minimum pass: 50% overall |

### Must-Know Facts Per Unit

#### Unit 1 — Hardware Fundamentals

- IPO Cycle: Input → Process → Output → Storage
- Hardware = physical. Software = instructions. Firmware = permanent software in ROM
- 5 computer types: Supercomputer > Mainframe > Minicomputer > Microcomputer > Embedded

#### Unit 2 — CPU

- ALU = calculations. CU = coordinates. Registers = fastest storage (inside CPU)
- Cache: L1 (fastest, smallest) → L2 → L3 (slowest, largest)
- Fetch-Decode-Execute cycle is the fundamental CPU operation
- Clock speed (GHz) × cores = overall processing capability
- Intel uses LGA sockets (pins on motherboard). AMD uses AM4/AM5 (pins on CPU)

#### Unit 3 — Memory

- RAM = volatile (data lost on power off). ROM = non-volatile
- Hierarchy (fastest to slowest): Registers > L1 Cache > L2 > L3 > RAM > SSD > HDD
- SRAM = fast (cache). DRAM = slower (main RAM). Flash = non-volatile (USB/SSD)
- DDR = Double Data Rate — transfers on both edges of clock cycle
- Virtual memory = HDD/SSD space used when RAM is full — much slower than real RAM

#### Unit 4 — Motherboard

- Motherboard installed FIRST, removed LAST
- Data bus = bidirectional. Address bus = unidirectional (CPU → memory)
- BIOS = old firmware. UEFI = modern replacement — supports drives >2TB
- POST = Power-On Self Test — 1 short beep = normal. Many beeps = error
- CMOS battery = small coin cell — stores BIOS settings — replace if date/time resets

## Unit 5 — Storage & PSU

- HDD = magnetic, moving parts, slow. SSD = flash, no moving parts, fast
- IDE needs master/slave jumpers. SATA does NOT need jumpers
- IDE = 2 drives per cable. SATA = 1 drive per cable
- NEVER defragment an SSD — it wears out flash memory needlessly
- PSU voltages: Yellow=+12V, Red=+5V, Orange=+3.3V, Black=Ground
- 80 PLUS rating = efficiency: Bronze < Silver < Gold < Platinum < Titanium

## Unit 6 — Software & OS

- Five OS functions: Process, Memory, File system, Device, User Interface management
- Open Source = free, modifiable (Linux). Proprietary = paid licence (Windows)
- Compiler = translates whole program at once. Interpreter = line by line
- Assembler = Assembly language → machine code (binary)

## Unit 7 — Troubleshooting

- 7 steps: Identify → Theory → Test → Plan → Fix → Verify → Document
- Preventive = before problems. Corrective = after problems. Predictive = monitor warning signs
- Always use ESD strap when handling internal components
- Start with SIMPLEST explanation first — check cables before replacing hardware

## Unit 8 — DOS & Networking

- Internal commands: DIR, CD, MD, RD, COPY, DEL, REN, TYPE, CLS, VER
- External commands: FORMAT (destroys data!), CHKDSK, FDISK, ATTRIB, XCOPY
- LAN = local area. WAN = wide area. Internet = the world's largest WAN
- Router = connects networks. Switch = connects devices within a LAN
- IP address = logical address (can change). MAC address = physical (permanent)
- WiFi: 802.11n=WiFi 4, 802.11ac=WiFi 5, 802.11ax=WiFi 6